Paleomagnetism of Truk Islands, Eastern Carolines and of Saipan, Marianas

M. FULLER, J. R. DUNN, G. GREEN, JIN-LU LIN, R. MCCABE, K. TONEY, AND I. WILLIAMS

Department of Geological Sciences, University of California, Santa Barbara, California 93106

Paleomagnetic results from Miocene flows from the Truk Islands reveal stable magnetization having a declination of +2° and an inclination of +6°, with an α_{95} of 6.8 and a k of 45.4. This is different from the present dipole field at the 95% confidence level. Ten sites are included in the analysis so that major departures from the true mean field direction due to secular variation should have been eliminated. The result gives an ancient latitude of 3°N with a ±3° confidence interval, which is consistent with proposed rotation about the Hawaiian hot spot pole at a rate of $1.3°/10^6$ years for the past 10 million years. The directions of magnetization of flows of Eocene and Miocene age from Saipan reveal a mean direction of magnetization rotated clockwise from the present field direction. Although there are fewer results available from Saipan than from Truk, the results appear to be recording some tectonic rotation and not simply secular variation. The Marianas limestone of Pleistocene age has a direction of magnetization concordant with the present field. These results from Truk and Saipan reveal that tectonic models can indeed be tested in island arc and oceanic island environments by paleomagnetic observations. It is also clear that the paleomagnetism of such islands suffers from inadequate numbers of available sites and difficulties in establishing tectonic control. The similarity of the rotation observed at Saipan and Guam strongly suggests that tectonic models of the Marianas must account for major rotation of the southern end of the Marianas since their formation.

Introduction

This paper describes paleomagnetic results from the Mariana-Philippine transect. Paleomagnetism may be particularly useful in unraveling the tectonics of the region; tectonic activity is recent and thus should be recorded by the natural remanent magnetization (NRM) of young rocks of the area, uncomplicated by later disturbing events. The absence of convincing sea floor magnetic anomalies with which to reconstruct tectonic history places additional importance on the paleomagnetic analysis. A difficulty in the application of paleomagnetism to the region is that there may well have been important east-west motion, which paleomagnetism does not detect owing to the longitudinal indeterminancy.

In this paper, results from Truk, in the Eastern Carolines, and Saipan, in the Marianas, are described. The former is of interest as a means of checking the predicted northward component of motion of the Pacific plate in the region. Saipan is primarily of interest as an aid to understanding the evolution of the Marianas Island arc. Already, *Larson et al.* [1975] have suggested that Guam, 150 miles to the southwest of Saipan, has been rotated since the Miocene and that it has maintained its present latitude throughout much of the Tertiary. In the discussion of their work on Guam they described an unsuccessful attempt to get paleomagnetic results from Saipan. They collected from only three sites, which gave inconsistent results. *Aoki* [1972] reported another attempt to do paleomagnetism on Saipan, and although the directions that were found were somewhat puzzling, he interpreted

Copyright American Geophysical Union

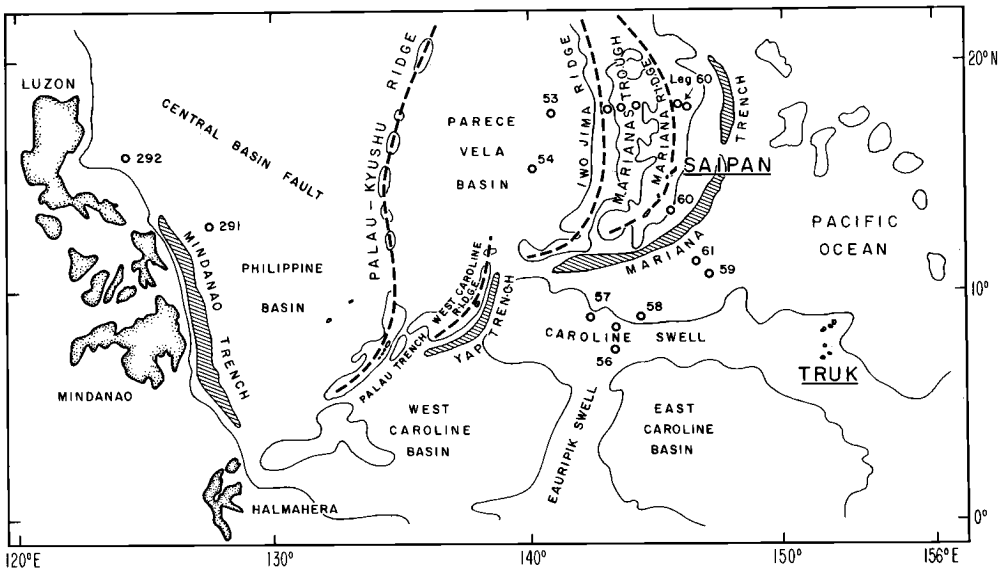


Fig. 1. Map of Mariana-Philippines transect.

at 30° clockwise rotation for the island of Saipan consistent with that of Guam. Our work on Saipan is aimed at extending these earlier studies, and eventually we hope to be able to establish when the rotation took place.

An additional source of paleomagnetic data is provided by the Deep-Sea Drilling Project (DSDP) holes for which detailed magnetic results are available. For example, *Louden* [1977] showed convincing evidence for northward motion of the west Philippine Basin. *Bleil* [1980] has observed similar results in the leg 60 holes, with the western sites showing marked translation northward. He interprets this as possible evidence of northwest-southeast spreading in the central part of the Mariana Trough. The paleomagnetic results from the Mariana Islands themselves could be important constraints on such models as those proposed by *Bleil* [1980] and those discussed by *Karig* [1978].

Truk Islands

Introduction

The Truk Islands are located at 7°N 152°E, about 700 miles southeast of Guam (Figure 1). The geology of the islands is described in U.S. Geological Survey professional paper 409 [*Stark and Hay*, 1963]. The islands are interpreted to be the remnants of a large shield volcano, which was active during the Miocene. We have obtained two K/Ar age determinations. One is for a flow on the north shore of Dublon Island, which gave an age of 11.2 ± 1.2 m.y. The other is for a flow from Moen, at sites MT5 on the Wichen River, about one half mile from its mouth. This was dated at 9.9 ± 0.7 m.y. The results from these islands may therefore permit an estimate of any north-south component of motion of the region for the past 10 m.y. Although the islands have luxuriant vegetation and typical tropical weathering is in evidence, there are several outcrops of fresh flows, mostly in coastal sea cliffs.

Paleomagnetic collections were made at the sites illustrated in Figure 2. It is our interpretation that at each site we have sampled an individual flow and that it is unlikely that the same flow was sampled at more than one site. The interpretation is based upon field relations and appears to be corroborated by the variation in magnetic properties and the NRM from site to site. Five to ten samples were used to define the direction of magnetization of the individual flows. The paleomagnetic data were for the most part of relatively high quality with small scatter.

Rock Magnetism

Rock magnetism studies have been carried out on selected samples, and the results are summarized in Table 1. They reveal that the dominant magnetic phase in the rocks has a low Curie point of between 150°C and 250°C. Optical microscopy revealed two predominant types of magnetic carriers. The low Curie point samples have coarse grain unexsolved titanomagnetite. The higher Curie point samples have much finer grained titanomaghemites.

Hysteresis measurements gave ratios of saturation remanence to saturation magnetization of 0.1–0.65, with MT 152 having the highest value. Its coercivity and remanent coercivity are also relatively high, and the ratio of H_{rc}:H_c is 1.3. These results strongly suggest that the magnetization of this sample is dominated by single domain carriers, which is consistent with the microscopic observation. The most multi-domain-like samples are MT 140 and

Copyright American Geophysical Union

10.1029/GM023p0235, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GM023p0235 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

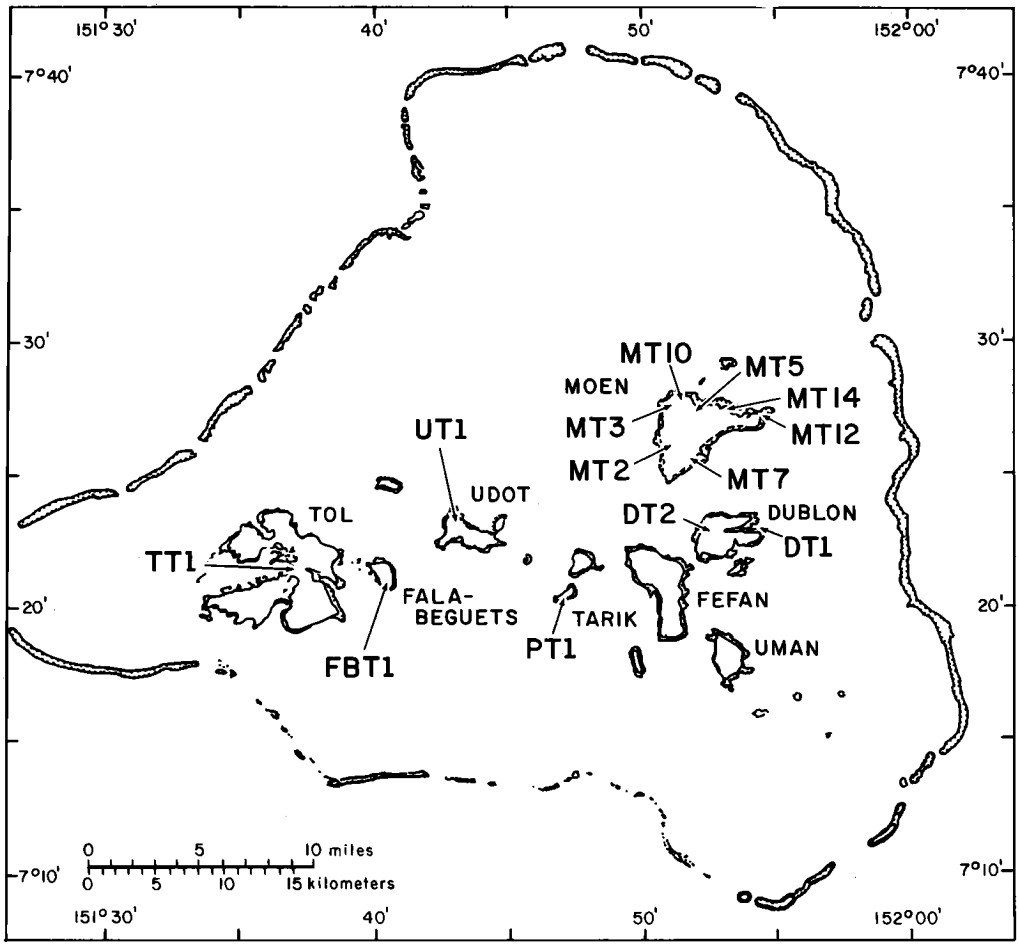


Fig. 2. Map of sites in the Truk Islands.

MT 143. This is consistent with microscopic observations described above.

The variation of alternating field (AF) demagnetization characteristics of the samples follow the trends seen in the hysteresis parameters. Thus MT 140 and MT 143 are the softest with mean destructive fields (MDF) of between 30 and 40 Oe for saturation isothermal remanence (IRM_s). In contrast, MT 152 is much harder with a mean destructive field for IRM_s of about 200 Oe. DT 101 is intermediate with a 75 Oe MDF for IRM_s. Acquisition of IRM follows a similar trend, with one half of the IRM_s value being acquired by 150 Oe by the softer samples and by about 200 Oe by the harder.

The acquisition of viscous magnetization (VRM) in a 5-Oe field was studied over a period of days. There was marked variation in the acquisition rates. For example, samples MT 120 and MT 123, which are similar to MT 140, acquired a magnetization of 10^{-3} G cm^3 g^{-1} within 100 hours. The stability of the VRM against AF demagnetization was markedly bimodal, with a large soft moment

TABLE 1. Rock Magnetism of Truk Samples

Samples	J_S, G cm^3 g^{-1}	J_R, G cm^3 g^{-1}	J_R/J_S	H_C, Oe	H_{RC}, Oe	H_{RC}/H_C	T_C, °C
MT 140	0.91	0.11	0.12	56	90	1.6	158
MT 143	0.78	0.11	0.14	56	80	1.4	···
MT 152	0.46	0.30	0.65	169	215	1.3	255
DT 101	0.31	0.09	0.29	84	155	1.8	183
DT 103	0.60	0.12	0.20	42	65	1.6	···

Copyright American Geophysical Union

10.1029/GM023p0235, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GM023p0235 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

whose MDF was about 20 Oe, but in addition, 3% was extremely hard, being untouched by 500-Oe demagnetization. The other samples tested, TT 104, TT 112, TT 113, MT 150, and MT 153, acquired very much less VRM, and it was not practicable to establish the AF demagnetization characteristics of the remanence acquired.

In summary, the magnetic properties of these samples from Truk Islands appear to reflect two different grain size carriers, which may also be intimately related to their oxidation state. This is comparable to results from DSDP basalts [e.g., *Day et al.*, 1978]. Samples such as MT 140 and MT 120 contain coarse grain titanomagnetites, which are magnetically soft and acquire VRM readily. In contrast, samples such as MT 152 and DT 101 from Dublon are magnetically harder and have a finer grain size, and in the extreme case of MT 152 the natural remanent magnetism is carried by single domain material that exhibits no important viscosity. We should pay special attention to any variation in paleomagnetic results that may be systematically related to these variations in magnetic characteristics.

Paleomagnetism

Samples were initially demagnetized to 200 Oe, in steps of 50 Oe to gain some impression of their AF demagnetization characteristics. This procedure appeared to isolate a primary stable NRM at most sites.

Figure 3 illustrates the range of AF demagnetization characteristics encountered. Figure 3*a* affords an example of a site at which the magnetization was soft and the samples viscous. However, its site mean direction is different from the present field direction at the 95% confidence level. Figure 3*b* shows the results from site MT5 where the samples are magnetically harder and less viscous. The magnetic carriers here are the fine grain titanomaghemites.

Thermal demagnetization has been carried out with selected samples, representing both the magnetically hard and soft material. Some of the results are shown in Figure 4. For the most part, agreement between AF and thermal demagnetization is good, being to within a few degrees. However, in some samples such as MT 217 the thermal demagnetization gave a steeper negative inclination than did the AF demagnetization, suggesting that the thermal demagnetization was more successful in eliminating magnetization acquired in the present shallow positive field direction.

The various site means for AF demagnetization to 200 Oe are given in Table 2 and Figure 5. All sites with the exception of site 11 give means that are significantly different from the present field at the 95% confidence levels. The inclinations are predominantly small and give a mean field direction with an inclination of +6°. The declination is +2.8°. It should be noted that site T 1 has not been included in this analysis since it clearly reflects not a tectonic event but a marked perturbation of the secular variation.

The interpretation of these results from Truk is relatively straightforward. Unfortunately, all sites are normally magnetized. It is therefore necessary to isolate an ancient primary NRM, which, in these young flows, is in a similar direction to the present geomagnetic field. Nevertheless, despite this difficulty and the fact that some of the samples are magnetically soft and viscous, it does appear that they have recorded a reliable ancient field direction. The sites give a mean direction of 002° declination and +6° inclination. The departure of the observed paleomagnetic field direction from the present dipole field is not correlated with the magnetic viscosity or softness of the samples. For example, site MT 2 exhibits one of the few negative inclination values, but the samples are magnetically soft and viscous. The interpretation of a record of paleomagnetic directions is confirmed by the similarity of the results from thermal demagnetization and AF demagnetization. The thermal demagnetization should be more effective at eliminating magnetic viscosity than AF demagnetization and yet both give similar results. This suggests that the viscosity, although present, is not strongly influencing the results. Nevertheless, it would probably be wise to regard the mean field direction observed as being weighted toward a minimum departure from the present field because of the viscosity of some of the samples.

Saipan

Introduction

Saipan, located at 15°N and 146°E, lies approximately 150 miles to the northeast of Guam, along the Marianas island arc. It consists of a volcanic core enveloped by limestones [*Cloud et al.*, 1956]. The possibility of obtaining paleomagnetic results for much of the Tertiary exists, but the combination of tropical weathering and limited outcrops makes paleomagnetism difficult on the island. We have sampled volcanic units of Eocene and Miocene age and also the ubiquitous limestones (Figure 6). We report here preliminary results from both volcanics and limestone.

The oldest volcanic material is the Sankakuyama formation of Eocene age. An age of 40.95 m.y. was obtained for an ash in the northern outcrop of the formation at the Bird Island lookout and of 41.5 m.y. for the flow below the ash (A. Meijer, private communication, 1980). The Sankakuyama is exposed in a small fault bounded block here, within which the bedding is shallowly inclined to the southeast. The formation is most extensively exposed in the central highlands to the east of Ogso Achaugau. Here again it is found in fault bounded blocks, but the dip here is highly variable. The Hagman formation is about 5 million years younger than the Sankakuyama (X. Meijer, private communication, 1980). Most of the exposures are very limited in extent and reveal badly weathered material. However, on the eastern flanks of the highlands, there are better if somewhat inaccessable sites. The Fina Sisu is the

Copyright American Geophysical Union

10.1029/GM023p0235. Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GM023p0235 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

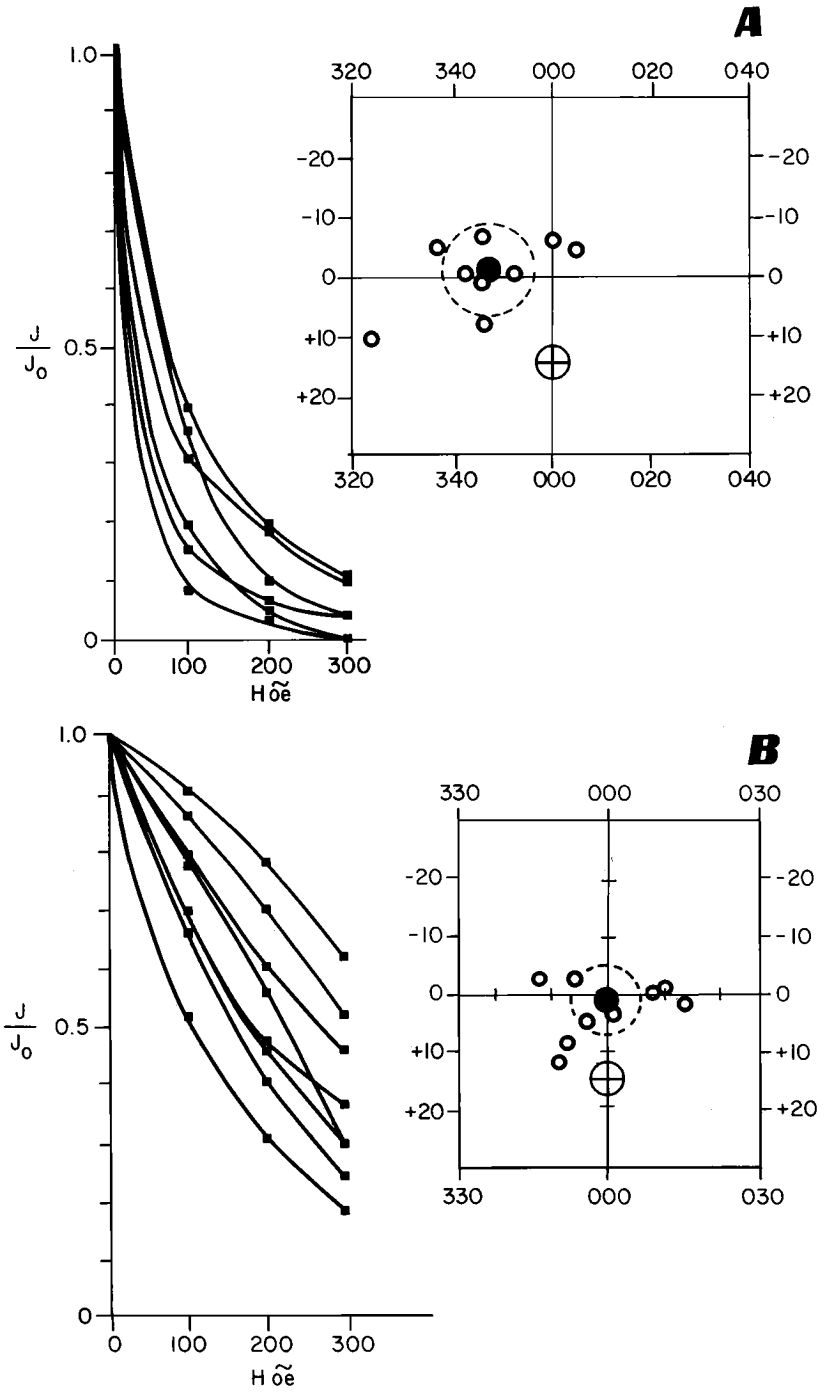


Fig. 3. Examples of paleomagnetic results from the Truk Islands. (*a*) Site MT2. (*b*) Site MT5.

youngest of the volcanic units (13 m.y.; X. Meijer, private communication, 1980). It is exposed in a single area in the southwest of the island, where it is flat lying.

The limestones of Saipan may eventually hold the key to determining the timing of any tectonic events that may be discovered, for they occur throughout geological time from the Eocene to the present. The Tagpochau is potentially the most interesting from this point of view because its age spans the Miocene and Oligocene. However, the NRM of this formation is very weak and multicomponent. At this time we have not succeeded in obtaining a satisfactory result for the formation. In contrast, the results

Copyright American Geophysical Union

10.1029/GM023p0235, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GM023p0235 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

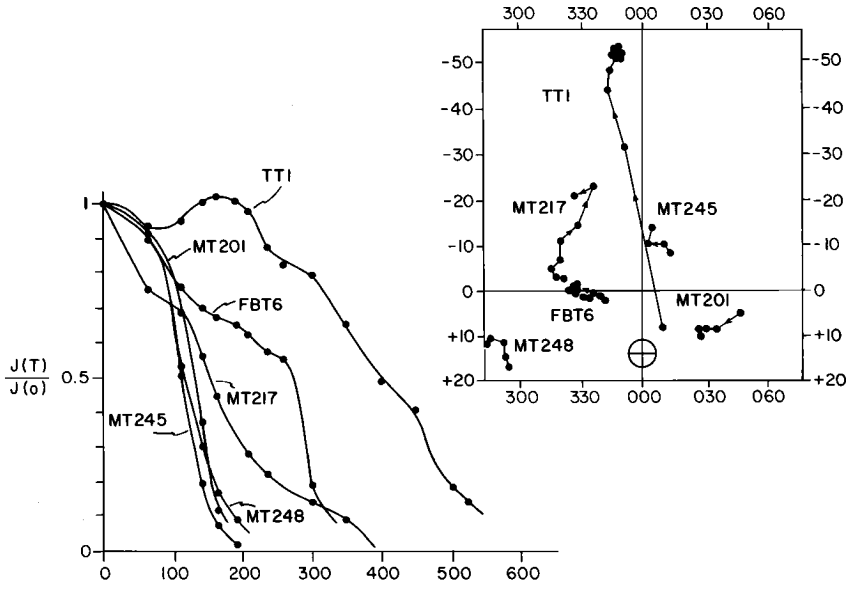


Fig. 4. Thermal demagnetization of samples from the Truk Islands.

from the Pleistocene Marianas limestone are more readily interpreted, and we report results from this formation.

In carrying out paleomagnetism on Saipan, one encounters some difficulty in establishing the tectonic corrections to be applied to paleomagnetic results [*Larson et al.*, 1975]. Often it is difficult to obtain a bedding attitude in the massive limestones, and the flows are frequently so limited in outcrop that there is again little field evidence of attitude. Even if one can observe the bedding it is not always clear whether the bedding dip is primary or tectonic. Fortunately, we have not been seriously affected by this problem since the majority of the sites for which we report results are either flat lying, or have such steep dips that they can safely be ascribed to tectonics. The northern exposure of the Sankakuyama at the Bird Island lookout is the one site where the difficulty is severe.

Rock Magnetism

The flow samples from Saipan are magnetically quite distinct from those from Truk (Table 3). Their Curie points are considerably higher. All three of the flow formations have Curie points in the region of 500°C. The hysteresis loops reveal intermediate magnetic hardness. The ratios of J_r/J_s range from 0.06 to 0.24, which suggests that the grain size of the magnetic carriers ranges upward from pseudo single domain to multidomain particles. The remanent magnetization characteristics are consistent with the hysteresis results, showing that the samples are indeed of intermediate hardness. It is notable that none of the samples show the very soft behavior seen in some of the Truk samples. The acquisition of VRM is moderate and need not be considered as an important contribution

TABLE 2. Paleomagnetism of Truk Samples

Site	Latitude	Longitude	Number of Samples	Inclination	Declination	α_{95}	K	Virtual Geomagnetic Pole		Δp	Δm
								Latitude	Longitude		
M2	7°21′N	151°51′E	9	−0.3°	348.2°	8.0	42.4	75.9°N	29.5°E	4.0	8.0
M3	7°28′N	151°51′E	7	8.3°	357.6°	6.1	100.1	86.3°N	12.4°E	3.1	6.1
M5	7°27′N	151°52′E	9	−3.7°	0.4°	7.9	43.4	80.6°N	31.0°W	3.9	7.9
M10	7°27′N	151°52′E	5	5.3°	14.9°	13.3	34.0	74.4°N	01.2°W	6.7	13.3
M11	7°27′N	151°53′E	6	15.5°	352.4°	11.9	32.5	82.5°N	65.3°E	6.3	12.2
D1	7°23′N	151°54′E	5	10.0°	24.3°	10.4	54.9	65.7°N	114.8°W	5.3	10.5
D2	7°23′N	151°52′E	8	4.6°	10.0°	10.5	28.6	79.0°N	93.2°W	5.3	10.6
U1	7°22′N	151°43′E	6	8.3°	6.8°	4.8	195.0	82.4°N	92.4°W	2.4	4.8
FB1	7°21′N	151°40′E	8	−0.4°	358.4°	4.4	156.7	82.1°N	16.9°W	2.2	4.4
T1	7°22′N	151°36′E	5	−47.4°	1.6°	9.2	70.2	53.9°N	30.9°W	7.7	11.9
P1	7°20′N	151°46′E	9	2.8°	359.5°	6.7	60.9	83.9°N	23.8°W	3.3	6.7
Mean	7°25′N	151°45′E	11	6.0°	2.8°	6.8	45.4	84.7°N	60.1°W	3.4	6.9

Copyright American Geophysical Union

10.1029/GM023p0235, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GM023p0235 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

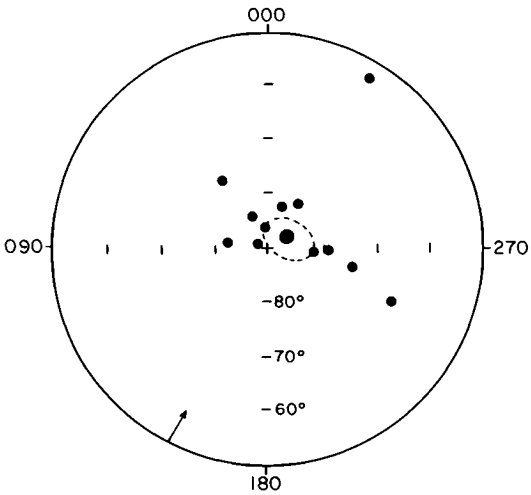


Fig. 5. Site means for the Truk Islands.

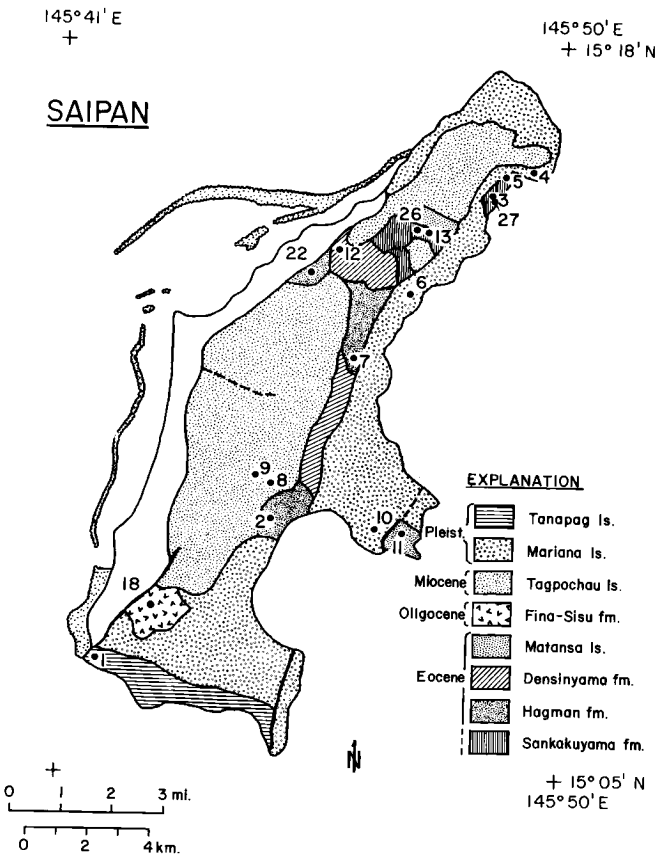


Fig. 6. Map of sites on Saipan.

to the NRM. To summarize, the magnetic characteristics of the samples suggest that they should be suitable for paleomagnetic studies. Further rock magnetism work on the flow samples will concentrate on attempting to establish the degree to which weathering has modified the carriers of NRM at the various sites.

The magnetic characteristics of the limestones are less readily established because they contain much less magnetic material than do the flows. However, AF demagnetization of the saturation IRM of the Marianas limestone from site 6 revealed that it was slightly softer than was the NRM. The remanent coercivity is 135. DC acquisition curves for IRM approached saturation at values of 1000–1500 Oe. Since the magnetization is not very viscous, these results suggest that it is primarily due to magnetite.

Paleomagnetism

The paleomagnetic results for Saipan are summarized in Table 4 and Figure 7. The latter gives the results for each site. In this preliminary discussion we have selected those sites for which a consistent direction of magnetization was obtained.

Sites 13 and 26 are in the central highland outcrop of the Sankakuyama and give high quality data showing no evidence of contamination by the present field (Figure 7*a*). The directions are unchanged by either AF or thermal demagnetization. The results from both sites have been corrected for bedding, which is evident at the sites in the form of a prominent foliation. The result is in agreement with locally mapped dip and strike of the flows. The tectonic correction makes very little difference to the declination of the NRM, but the inclination is essentially determined by the tectonic correction. Both corrected and uncorrected directions are shown in the figure. Unfortunately, the similar dips at both sites preclude a convincing fold test.

To the north at the Bird Island lookout there is a welded tuff (site 3) and a flow (site 27) in the Sankakuyama. These exhibit a stable direction upon AF and thermal demagnetization, which is similar to the present dipole field (Figure 7*b*). The bedding is clearly seen in the ash, and it

TABLE 3. Rock Magnetism of Saipan Samples

Formation	Site	J_S, G cm^3 g^{-1}	J_R, G cm^3 g^{-1}	J_R/J_S	H_C, Oe	H_{RC}, Oe	H_{RC}/H_C	T_C, C°
Fina Sisu	18	2.1	0.36	0.17	75	125	1.7	500
Hagman	2	0.58	0.1	0.17	98	245	2.5	500
Sankakuyama	13	0.12	0.028	0.24	146	292	2.0	525
Marianas limestone	6	0.005				135		
		0.018				135		

Copyright American Geophysical Union

10.1029/GM023p0235, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GM023p0235 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

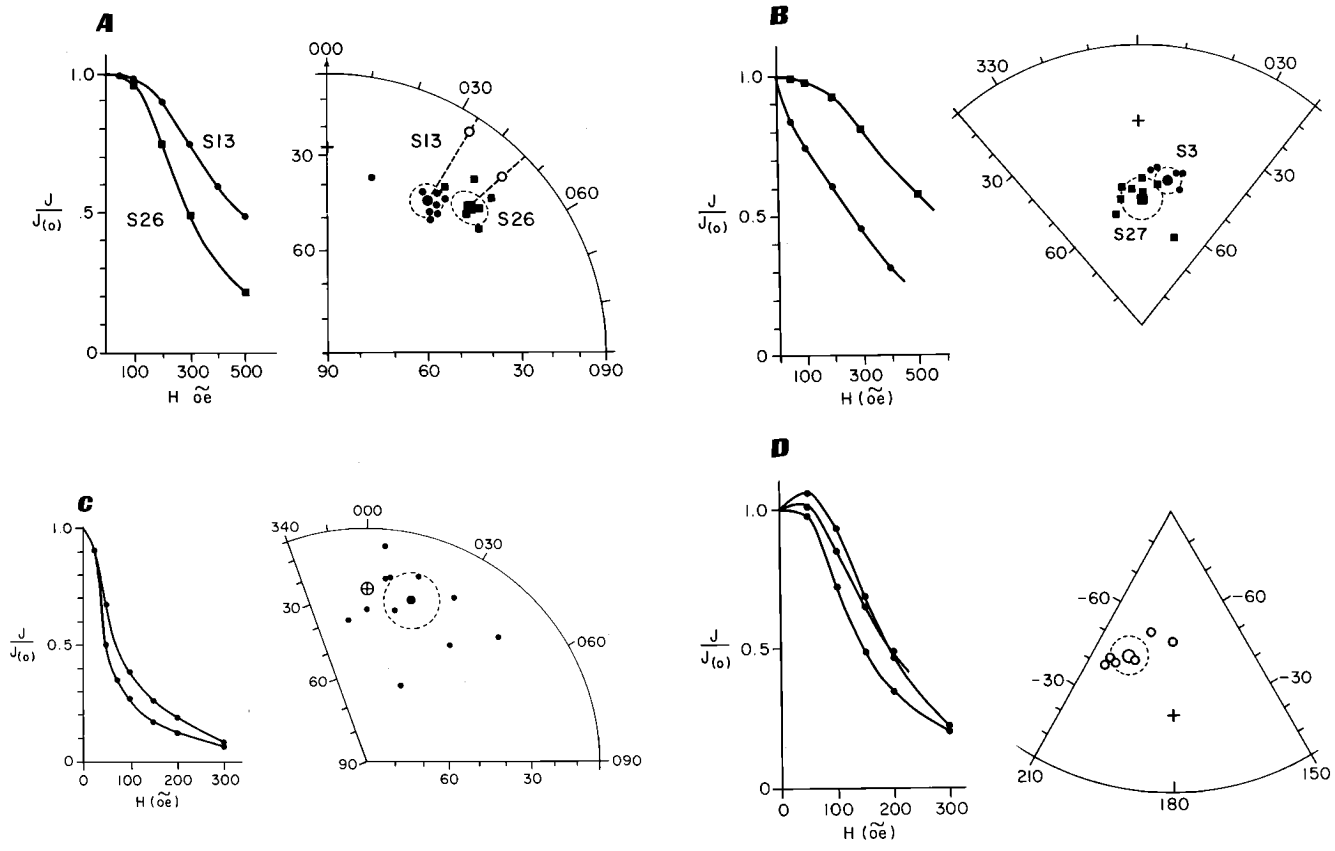


Fig. 7. Paleomagnetic results from Saipan. (*a*) Sankakuyama formation, sites 13 and 26. (*b*) Sankakuyama formation, sites 3 and 27. (*c*) Hagman formation, site 11. (*d*) Fina Sisu formation, site 18. (*e*) Marianas limestone, site 6.

is shallowly inclined to the southeast. A difficulty in the interpretation of these results is that the uncorrected direction of magnetization is very similar to the present field direction. However, the stability of the NRM against AF and thermal demagnetization suggests that the direction is unlikely to be secondary.

The Hagman flows have not given high quality uncontaminated NRM directions. However, the results for site 11 in the southeast of the island near Hagman point are described here. The material is variably weathered. The attitude of the flows at this site is quite apparent and assumed to be tectonically controlled. The directions of mag-

TABLE 4. Paleomagnetism of Saipan Samples

Formation	Site	Latitude	Longitude	Number of Samples	Inclination	Declination	α_{95}	K	Virtual Geomagnetic Pole Latitude	Virtual Geomagnetic Pole Longitude	Δp	Δm
Marianas limestone	6	15°14′N	145°47′E	10	26.0°	358.7°	4.9	97.6	88.1°N	10.7°E	2.9	5.3
Fina sisu	18	15°8′N	145°42′E	6	−44.9°	196.0°	6.4	111.4	71.2°S	15.9°E	5.1	8.0
Hagman	11	15°14′N	145°46′E	11	29.8°	15.4°	11.4	17.1	75.2°N	129.2°W	7.0	12.6
Sankakuyama	13	15°14′N	145°47′E	8	35.9°	34.9°	4.9	126.8	56.4°N	137.5°W	3.3	5.7
Sankakuyama	26	15°14′N	145°47′E	7	27.3°	45.8°	5.2	134.1	45.8°N	129.5°W	3.1	5.7
Sankakuyama	3	15°16′N	145°49′E	5	45.6°	11.0°	4.4	30.5	74.2°N	175.4°W	3.6	5.6
Sankakuyama	27	15°16′N	145°49′E	8	53.3°	357.5°	5.6	100.1	71.1°N	139.5°E	5.4	7.7
Mean of Hagman and Sankakuyama sites 3, 11, 13, 26, and 27				5	39.6°	22.9°	17.7	19.7	67.2°N	146.3°W	12.7	21.2

Copyright American Geophysical Union

10.1029/GM023p0235, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GM023p0235 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

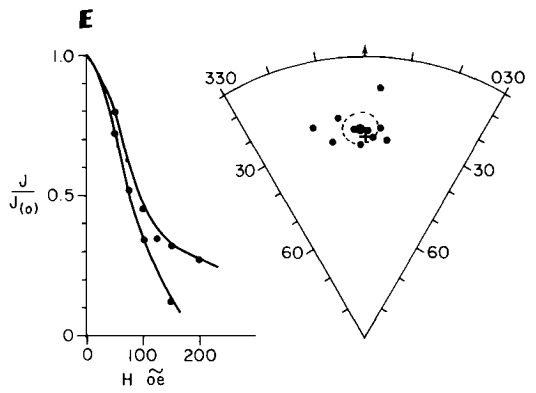


Fig. 7. (continued)

netization are shown in Figure 7*c*. The mean declination is rotated by about 15° with respect to the present field, but the α_{95} circle includes the present field direction. Our interpretation of these results is that the primary NRM is substantially rotated with respect to the present field but that the secondary magnetization acquired during weathering in the present field is so strong that it precludes accurate determination of the primary NRM.

The Fina Sisu is now exposed only in tunnels in the southwest of the island. Immediately after the World War II, outcrops were evidentally more numerous and the material is described by *Cloud et al.* [1956] in regions that have now reverted to jungle. The flows are flat lying, and the direction of magnetization is reversed (Figure 7*d*). The mean direction has a declination of 196 and an inclination of −45°. This corresponds again to a clockwise rotation of NRM with respect to the present field.

Of the various limestone sites studied, site 6, in the Marianas limestone of Pleistocene age, gives the most consistent direction of magnetization. The site is on the east coast of the island near to the Unai Fahung. The beds dip shallowly to the east. The directions of NRM after AF demagnetization are shown in Figure 7*e*. Applying a tectonic correction brings the direction of magnetization into coincidence with the present local dipole field. Without the correction, they have a small negative declination. This result suggests that the samples are revealing a primary NRM that has survived a tectonic tilting recorded by the shallow dip of these limestones.

Thermal demagnetization was carried out with samples from the Sankakuyama and Fina Sisu flows and the Marianas limestone. Samples from the Sankakuyama at site 13 and the Fina Sisu at site 18 showed no significant changes upon thermal demagnetization. They were thus entirely consistent with the AF demagnetization studies, which had shown a single stable direction of NRM. A sample from the flow at site 27 in the Sankakuyama showed a small increase in declination upon thermal demagnetization but gave a final result within the α_{95} of the site mean determined from the AF demagnetization. The two limestone samples from the Marianas limestone at site 6 exhibited a slight shallowing of inclination upon thermal demagnetization and opposite minor trends in declination. The results of thermal demagnetization of all of these samples are therefore consistent with those from AF demagnetization.

We have reported results from 7 sites on Saipan. Of these, two of the oldest have a substantial clockwise rotation of the NRM. The other two sites from the Sankakuyama have stable NRM in the same direction as the present field. The Hagman is represented by a single site that gives a mean direction having an inclination of +15° and a declination of +30°, which is similar to the direction at sites 13 and 26 in the Sankakuyama. The mean direction for these five sites has a declination of +23° and an inclination of 40°. However, the α_{95} is 11° and the k only 20. The youngest of the volcanics, the Fina Sisu, has a reversed NRM with a declination of 196° and an inclination of −45°. Finally, the Marianas limestone has an NRM parallel to the present field direction. The NRM of the older sites is thus rotated with respect to the present field. At this point the rotation is not accurately determined, but is probably not more than 40° or less than about 10°. The inclinations of the magnetization at the various sites is either coincident with the present field or steeper than it.

Discussion

The use of paleomagnetic data in tectonic analyses presumes that the paleomagnetic data give a good estimate of the mean dipole field direction at the site and that the effects of secular variation have been eliminated. In many sediments and intrusive igneous rocks the NRM represents an average field value owing to the time taken for the sample to acquire its magnetization. Unfortunately, in lavas, such as those that provide a major part of the present data from Truk and Saipan, this is not the case. The flows essentially record spot readings of the field and hence record its secular variation. The problem is then to establish how many flow samples are required to give a good mean field estimate. The approach that we have taken is to determine, from a Fisher analysis, the precision parameter k, which is a measure of the variance of the recorded directions. We then compare it with other available secular variation data and the various secular variation models.

The precision factor k determined from the Truk sites is 54 with the TT1 site excluded. This value is somewhat high in comparison with the standard models of secular variation and the data from the Galapagos, although not with the Hawaiian data [e.g., *McElhinny and Merrill*, 1975]. These results suggest that the ten sites sampled in the Truk islands are sufficient to eliminate significant effects of secular variation. The difference between the paleomagnetically recorded inclination of +6° and the present dipole field inclination of +15° is therefore to be interpreted as a reflection of motion of the Truk islands with respect to the magnetic pole. It is then possible to compare this northward translation with that predicted by the motion

Copyright American Geophysical Union

10.1029/GM023p0235, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GM023p0235 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

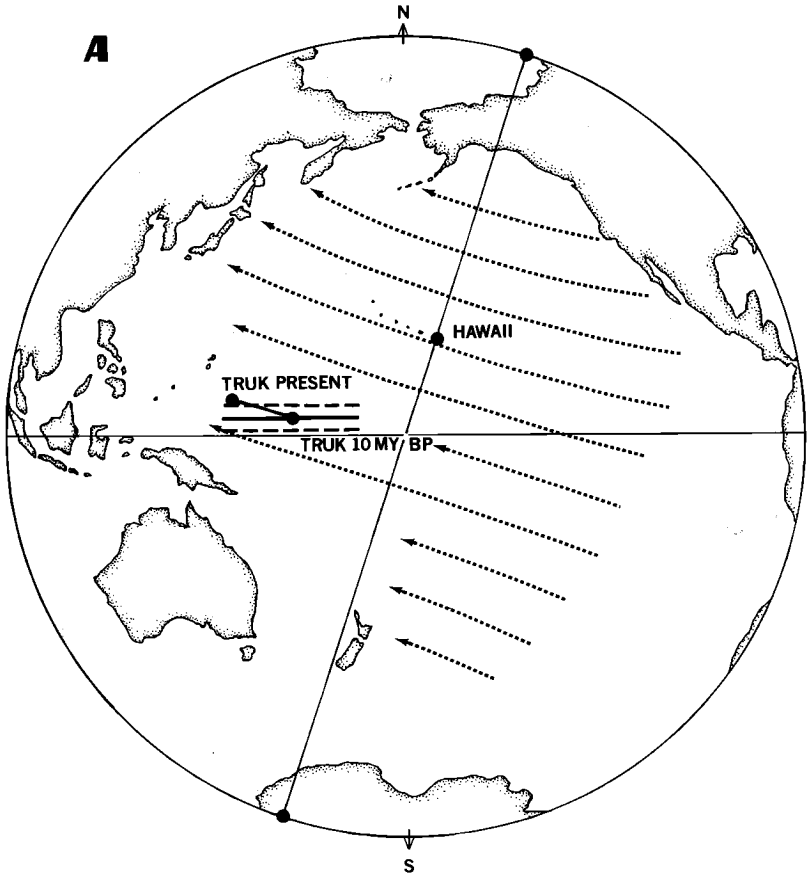


Fig. 8. Tectonic implications of paleomagnetic results from Truk and Saipan. (*a*) Paleomagnetism of Truk and the motion of the Pacific plate. (*b*) Paleomagnetism of Saipan and the deformation of the Marianas Island Arc.

of the Pacific plate about the Hawaiian hot spot rotation pole at a rate of 1.3°/m.y. as was suggested by *Clague and Jarrard* [1973]. Figure 8 illustrates the situation using an equal area projection with the hot spot rotation pole on the primitive as shown. The predicted value of the northward component of motion is then seen to be 3°. This value is that obtained from the paleomagnetic results so that the product values fall within the 95% confidence interval of the paleomagnetically determined latitude. Our study is thus consistent with this suggested motion of the Pacific plate for the last 10 m.y.

The results from Saipan bear upon the history of the Marianas Island Arc. They are, as we noted above, not yet conclusive, but they do suggest that between about 40 m.y. ago, when the Sankakuyama volcanics were extruded, and the Pleistocene, when the Marianas limestones were formed, the island suffered a rotation of between 10° and 40°. This result is similar to the result of *Aoki* [1972]. However, he, like we, found sites with aberrant directions of NRM. His evidence for rotation is principally at a single site in the Hagman, so that he was also unable to make a very strong case for the rotation. Thus the work on Saipan and on Guam [*Larson et al.*, 1975] suggests that the southern part of the arc has been rotated since formation, but additional work is required on each island to determine the rotation accurately. Neither island gives sufficiently accurately determined inclinations to permit changes in latitude to be specified, but they appear to have been minor if any. This result is in marked conflict with the result from the nearby DSDP sediments described by *Bleil* [1980]. They also appear to be in conflict with the results of *Louden* [1977], although these data sets may be reconcilable in terms of certain models of the evolution of the region [*Karig*, 1978].

We plan to continue our work on Saipan to determine the rotation more accurately, to date the period of rotation, and to attempt to get better information concerning the ancient latitude of the island of Saipan.

Acknowledgments. The authors wish to express their gratitude to the Southeast Asia Tectonics and Resources (Seatar) organization committee who made available a

Copyright American Geophysical Union

10.1029/GM023p0235, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GM023p0235 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

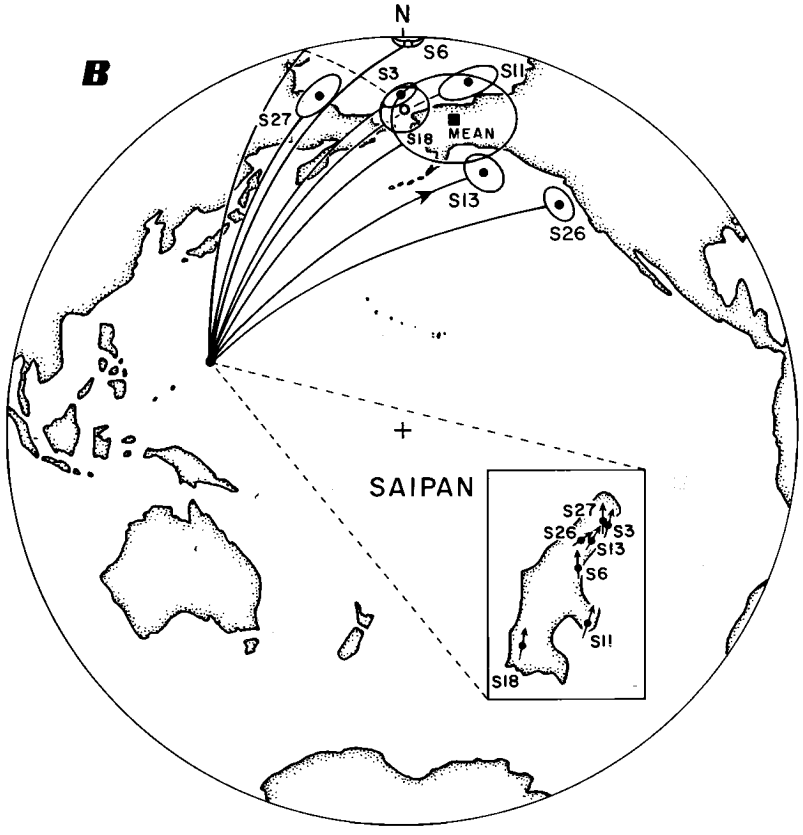


Fig. 8. (continued)

travel grant that made the initial fieldwork possible and to the National Science Foundation for subsequent support through NSF OCE23920. We are also grateful to B. Davis and M. Osterhoudt for help in measurements of Curie points and hysteresis loops, respectively. We thank the members of the departmental office for typing the manuscript and D. Crouch for the diagrams.

References

Aoki, Y., A post depositional magnetization mechanism of sediments and its application to geophysical problems: Paleosecular variation in Japan, Ph.D. thesis, University of Tokyo, 1972.

Bleil, U., Paleomagnetism of DSDP leg 60 sediments and igneous rocks from the Mariana region, in *Initial Reports of the Deep-Sea Drilling Project*, vol. 60, U.S. Government Printing Office, Washington, D. C., 1980.

Clague, D. A., and R. D. Jarrard, Tertiary Pacific plate motion deduced from the Hawaiian-Emperor chain, *Geol. Soc. Amer. Bull., 84,* 1135-1154, 1973.

Cloud, P. E., R. G. Schmidt, and H. W. Burke, Geology of Saipan, Mariana Islands, Part 1: General geology, *Prof. Pap. 280,* U.S. Geol. Surv., 1956.

Day, R., S. Halgedahl, M. Steiner, K. Kobayashi, T. Furuta, T. Ishii, and A. Faller, Magnetic properties of basalts from DSDP leg 49, in *Initial Reports of the Deep-Sea Drilling Project*, vol. 49, U.S. Government Printing Office, Washington, D. C., 1978.

Karig, D. E., R. N. Anderson, and L. D. Bibee, Characteristics of back arc spreading in the Mariana trough, *J. Geophys. Res., 83,* 1212-1226, 1978.

Larson, E. E., R. L. Reynolds, M. Ozima, Y. Aoki, H. Kinoshita, S. Zasshu, N. Kawai, T. Nakajima, K. Hirooka, R. Merrill, and S. Levi, Paleomagnetism of Miocene volcanic rocks from Guam and the curvature of the southern Mariana island arc, *Geol. Soc. Amer. Bull., 86,* 346-650, 1975.

Louden, K., Paleomagnetism of DSDP sediments, phase shifting of magnetic anomalies and rotations of the West Philippine basin, *J. Geophys. Res., 82,* 2989-3002, 1977.

McElhinny, M. W., and R. T. Merrill, Geomagnetic secular variation over the past 5 m.y., *Rev. Geophys. Space Phys., 13,* 687-708, 1975.

Stark, J. T., and R. L. Hay, Geology and Petrology of the volcanic rocks of Truk Islands East Caroline Islands, *Prof. Pap. 409,* p. 409, U.S. Geol. Surv., 1963.

Copyright American Geophysical Union

10.1029/GM023p0235, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GM023p0235 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License